NORMALIZATION OF DATABASE

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STUDY OBJECTIVES

- Understand what normalization is and what role it plays in database design
- Learn about the normal forms 1NF, 2NF, 3NF
- Identify how normal forms can be transformed from lower normal forms to higher normal forms
- Understand normalization and E-R modeling are used concurrently to produce a good database design
- Understand some situations require denormalization to generate information efficiently

WHY NORMALIZATION?

- To produce well-structured relations
  - Any relational database should contain minimal data redundancy and allows users to insert, delete, and update rows without causing data inconsistencies (anomalies).
  - Goal of Normalization: producing well-structured relations by eliminating anomalies

TYPE OF ANOMALIES

- Update (Modification) Anomaly
  - Changing data in a row forces changes to other rows because of duplication
- Deletion Anomaly
  - Deleting rows may cause a loss of data that would be needed for other future rows
- Insertion Anomaly
  - Adding new rows forces user to create duplicate data

EACH ANOMALY EXAMPLES

CONSIDER THE FOLLOWING TABLE THAT STORES DATA ABOUT AUTO PARTS AND SUPPLIERS. THIS SEEMINGLY HARMLESS TABLE CONTAINS POTENTIAL PROBLEMS.

Part#	Description	Supplier	Address	City	State
100	Coil	Dynar	45 Eastern Ave.	Denver	CO
101	Muffler	GlassCo	1638 S. Front	Seattle	WA
102	Wheel Cover	A1 Auto	7441 E. 4th Street	Detroit	MI
103	Battery	Dynar	45 Eastern Ave.	Denver	CO
104	Radiator	United Parts	346 Taylor Drive	Austin	TX
105	Manifold	GlassCo	1638 S. Front	Seattle	WA
106	Converter	GlassCo	1638 S. Front	Seattle	WA

Suppose, we need to add another part.

107 Tail Pipe GlassCo 1638 S. Front Seattle WA

UPDATE ANOMALY

WHAT IF GLASSCO MOVES TO OLYMPIA? HOW MANY ROWS HAVE TO BE CHANGED IN ORDER TO ENSURE THAT THE NEW ADDRESS IS RECORDED – ADDRESS AND CITY

Part#	Description	Supplier	Address	City	State
100	Coil	Dynar	45 Eastern Ave.	Denver	CO
101	*Muffler*	*GlassCo*	*1638 S. Front*	*Seattle*	*WA*
102	Wheel Cover	A1 Auto	7441 E. 4th Street	Detroit	MI
103	Battery	Dynar	45 Estern Ave.	Denver	CO
104	Radiator	United Parts	346 Taylor Drive	Austin	TX
105	*Manifold*	*GlassCo*	*1638 S. Front*	*Seattle*	*WA*
106	*Converter*	*GlassCo*	*1638 S. Front*	*Seattle*	*WA*
107	*Tail Pipe*	*GlassCo*	*1638 S. Front*	*Seattle*	*WA*

DELETION ANOMALY

SUPPOSE YOU NO LONGER CARRIES PART NUMBER 102 AND DECIDE TO DELETE THAT ROW FROM THE TABLE?

Part#	Description	Supplier	Address	City	State
100	Coil	Dynar	45 Eastern Ave.	Denver	CO
101	Muffler	GlassCo	1638 S. Front	Seattle	WA
102	*Wheel Cover*	***A1 Auto***	*7441 E. 4th Street*	*Detroit*	*MI*
103	Battery	Dynar	45 Estern Ave.	Denver	CO
104	Radiator	United Parts	346 Taylor Drive	Austin	TX
105	Manifold	GlassCo	1638 S. Front	Seattle	WA
106	Converter	GlassCo	1638 S. Front	Seattle	WA
107	Tail Pipe	GlassCo	1638 S. Front	Seattle	WA

NOW, LOOKING AT THE REMAINING DATA BELOW, WHAT IS THE ADDRESS OF A1 AUTO? - SUPPLIER (A1 AUTO) ADDRESS MUST BE DELETED AS WELL

Part#	Description	Supplier	Address	City	State
100	Coil	Dynar	45 Eastern Ave.	Denver	CO
101	Muffler	GlassCo	1638 S. Front	Seattle	WA
103	Battery	Dynar	45 Estern Ave.	Denver	CO
104	Radiator	United Parts	346 Taylor Drive	Austin	TX
105	Manifold	GlassCo	1638 S. Front	Seattle	WA
106	Converter	GlassCo	1638 S. Front	Seattle	WA
107	Tail Pipe	GlassCo	1638 S. Front	Seattle	WA

INSERTION ANOMALY

NEXT, YOU WANT TO ADD A NEW SUPPLIER – "CARPARTS." BUT YOU HAVE NOT YET ORDERED PARTS FROM THAT SUPPLIER – NO PK AND DESCRIPTION

Part#	Description	Supplier	Address	City	State
100	Coil	Dynar	45 Eastern Ave.	Denver	CO
101	Muffler	GlassCo	1638 S. Front	Seattle	WA
103	Battery	Dynar	45 Estern Ave.	Denver	CO
104	Radiator	United Parts	346 Taylor Drive	Austin	TX
105	Manifold	GlassCo	1638 S. Front	Seattle	WA
106	Converter	GlassCo	1638 S. Front	Seattle	WA
107	Tail Pipe	GlassCo	1638 S. Front	Seattle	WA
???	*????????*	*CarParts*	*101 Mariposa*	*Orlando*	*FL*

UTILIZING FUNCTIONAL DEPENDENCY THEORY

- As a solution for taking care of Anomalies
- Normalization is based on functional dependencies.
- Functional Dependency: The value of one attribute determines the value of another attribute
- Notation: $\Rightarrow$ (arrow)
- $A \rightarrow B$ when value of A (of a valid instance) defines the value of B (B is functionally dependent upon A).
  - SSN defines Name, Address (not vice versa)
- A is the *determinant* in a functional dependency

EXAMPLE OF FUNCTIONAL DEPENDENCY

- SSN -> Name, Birth-date, Address
  - VIN -> Make, Model, Color
  - ISBN -> Title, Author
- Not acceptable dependencies
  - Partial dependency
  - Transitive dependency
  - Hidden dependency

FIRST NORMAL FORM (1NF)

- To be in First Normal Form (1NF),
  - Each column must contain only a single value
  - Repeating groups of records (redundancy) must be eliminated
    - Eliminate duplicative columns from the same table.
  - There must not be a composite and a multi-valued attributes.
    - Transformation from model to relation

1NF Example

Unnormalized Table

PK

OrderNum	OrderDate	PartNum	NumOrdered
21608	10/20/2003	AT94	11
21610	10/20/2003	DR93 DW11	1 1
21613	10/21/2003	KL62	4
21614	10/21/2003	KT03	2
21617	10/23/2003	BV06 CD52	2 4
21619	10/23/2003	DR93	1
21623	10/23/2003	KV29	2

1NF Example (con't.)

Conversion to 1NF

PK

OrderNum	OrderDate	PartNum	NumOrdered
21608	10/20/2003	AT94	11
21610	10/20/2003	DR93	1
21610	10/20/2003	DW11	1
21613	10/21/2003	KL62	4
21614	10/21/2003	KT03	2
21617	10/23/2003	BV06	2
21617	10/23/2003	CD52	4
21619	10/23/2003	DR93	1
21623	10/23/2003	KV29	2

ANOTHER 1NF EXAMPLE

PK

Cust_ID	L_Name	F_Name	Address
104	Suchecki	Ray	123 Pond Hill Road, Detroit, MI, 48161

PK

Cust_ID	SalesRep_Name	Rep_Office	Order_1	Order_2	Order_3
1022	Jones	412	10	14	19

SECOND NORMAL FORM

- In order to be in 2NF, a relation must be in 1NF and a relation must **not have any partial dependencies.**
  - Any attributes must not be dependent on a portion of primary key.
- The other way to understand 2NF is that each non-key attribute (not a part of PK) in the relation must be functionally dependent upon the primary key.

2NF Example

PK PK

OrderNum	OrderDate	PartNum	Description	NumOrdered	QuotedPrice
21608	10/20/2003	AT94	Iron	11	$21.95
21610	10/20/2003	DR93	Gas Range	1	$495.00
21610	10/20/2003	DW11	Washer	1	$399.99
21613	10/21/2003	KL62	Dryer	4	$329.95
21614	10/21/2003	KT03	Dishwasher	2	$595.00
21617	10/23/2003	BV06	Home Gym	2	$794.95
21617	10/23/2003	CD52	Microwave Oven	4	$150.00
21619	10/23/2003	DR93	Gas Range	1	$495.00
21623	10/23/2003	KV29	Treadmill	2	$1290.00

Each arrow shows partial dependency

OrderNum, PartNum → NumOrdered, QuotedPrice

OrderNum → OrderDate / PartNum → Description

2NF Example

PK

OrderNum	OrderDate
21608	10/20/2003
21610	10/20/2003
21613	10/21/2003
21614	10/21/2003
21617	10/23/2003
21619	10/23/2003
21623	10/23/2003

PK

PartNum	Description
AT94	Iron
BV06	Home Gym
CD52	Microwave Oven
DL71	Cordless Drill
DR93	Gas Range
DW11	Washer
FD21	Stand Mixer
KL62	Dryer
KT03	Dishwasher
KV29	Treadmill

PK PK

OrderNum	PartNum	NumOrdered	QuotedPrice
21608	AT94	11	$21.95
21610	DR93	1	$495.00
21610	DW11	1	$399.99
21613	KL62	4	$329.95
21614	KT03	2	$595.00
21617	BV06	2	$794.95
21617	CD52	4	$150.00
21619	DR93	1	$495.00
21623	KV29	2	$1290.00

THIRD NORMAL FORM

- In order to be in Third Normal Form, a relation must first fulfill the requirements to be in 2NF.
- Additionally, all attributes that are not dependent upon the primary key must be eliminated. In other words, there should be no transitive dependencies.
  - remove columns that are not dependent upon the primary key.

EXAMPLE OF 3NF

PK: Cust_ID

SALES

Cust_ID	Name	Salesperson	Region
8023	Anderson	Smith	South
9167	Bancroft	Hicks	West
7924	Hobbs	Smith	South
6837	Tucker	Hernandez	East
8596	Eckersley	Hicks	West
7018	Arnold	Faulb	North

TRANSITIVE DEPENDENCY

- All attributes are functionally dependent on Cust_ID.
  - Cust_ID → Name, Salesperson
- However, there is a transitive dependency.
  - <u>Region is functionally dependent on Salesperson</u> but Salesperson is not a "Determinants."
  - Salesperson → Region

SALES

Cust_ID	Name	Salesperson	Region
8023	Anderson	Smith	South
9167	Bancroft	Hicks	West
7924	Hobbs	Smith	South
6837	Tucker	Hernandez	East
8596	Eckersley	Hicks	West
7018	Arnold	Faulb	North

- What kind of anomaly will happen if the company needs to add a new salesperson (John Doe) to the North region right away?
- What kind of anomaly will happen tomorrow if the company deletes the customer number 6837 from the table today?
- What kind of anomaly will happen if Smith (salesperson) must be reassigned to the East region right away?

Decomposing the SALES relation

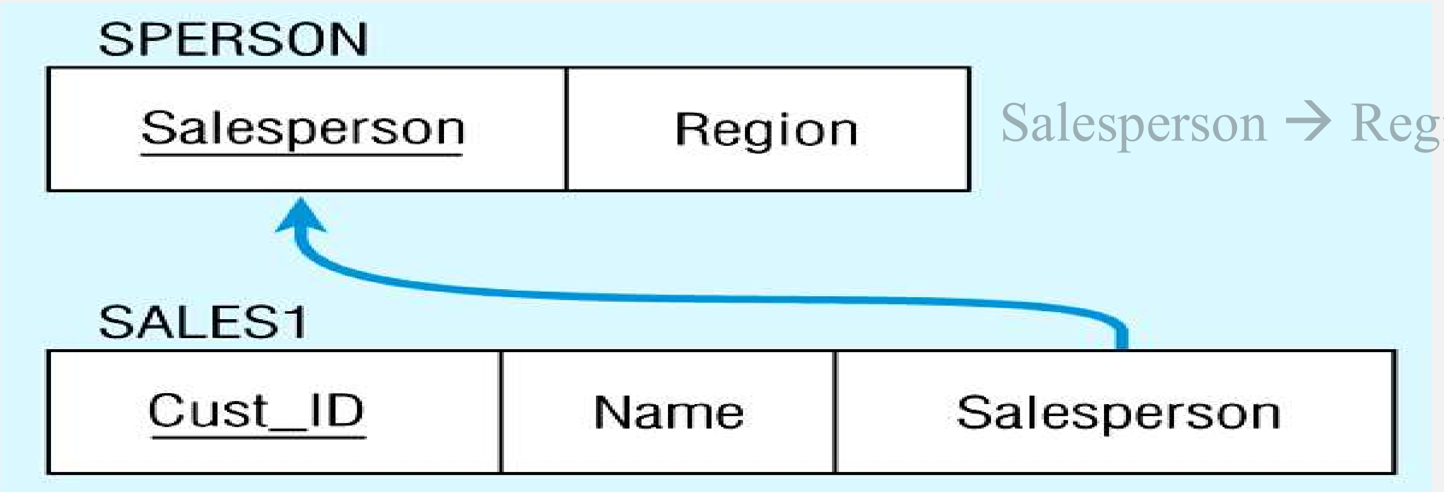
SALES1 (FK: Salesperson)

PK Cust_ID	Name	Salesperson
8023	Anderson	Smith
9167	Bancroft	Hicks
7924	Hobbs	Smith
6837	Tucker	Hernandez
8596	Eckersley	Hicks
7018	Arnold	Faulb

SPERSON PK

Salesperson	Region
Smith	South
Hicks	West
Hernandez	East
Faulb	North

Relations in 3NF



Salesperson → Region

CustID → Name and CustID → Salesperson

Now, there are no transitive dependencies…
Both relations are in 3rd NF

STOP HERE!

- Very interesting logic.
- If you are interested in learning this topic, please let me know.
- We will explore together in detail!

NORMALIZATION AND DATABASE DESIGN

- Normalization should be part of the design process
  - Unnormalized:
    - Data updates less efficient
    - Indexing more cumbersome
- E-R Diagram provides macro view
- Normalization provides micro view of entities
  - Focuses on characteristics of specific entities
  - May yield additional entities
- Generally, most database designers do not attempt to implement anything higher than Third Normal Form or Boyce-Codd Normal Form.